

ECOCIVIX AI

03 — 3D Experience & Animation Specification

TEAM VELOCITY • HackYatra × EDM 3.0 • PRE-HACKATHON PLANNING

Purpose: Define how 3D, motion, transitions, micro-interactions and visual effects should be used throughout ECOCIVIX AI. Motion must communicate intelligence, connection, environmental awareness, civic activity, progress, state changes and feedback rather than exist as decoration.

Status: PRE-HACKATHON DRAFT / PLANNING SPECIFICATION

Approval authority: Garv Nain

Experience North Star

- A living digital intelligence layer over a real community.
- The visual language combines civic infrastructure, environmental systems, community signals and AI intelligence.
- Premium, intelligent, calm, precise, futuristic, trustworthy, environmental and human-centered.

Motion Principles

- Purpose before decoration.
- Level 1 core motion: buttons, forms, loading, status and transitions.
- Level 2 product motion: AI analysis, timeline, dashboard, map and data visualization.
- Level 3 showcase motion: hero 3D, advanced environment and intelligence-core visuals.
- If time is limited: Level 1 → Level 2 → Level 3.

3D Strategy

- Use primarily on the landing page, product storytelling, intelligence visualization and selected community-intelligence moments.
- Do not require 3D for login, signup, issue forms, issue tracking, admin operations or accessibility-critical information.
- Core UI must remain functional if WebGL/3D fails.

Hero Experience

- Central ECOCIVIX visual system may represent community signals, civic infrastructure, environmental data, safety relevance and AI intelligence.
- Possible elements: floating data nodes, connected points, geographic structures, subtle particles, central intelligence core and flowing data lines.
- Hero motion: entry → connection → intelligence activation → headline/CTA → calm idle state.

AI Analysis Motion

- Represent UI-level stages: Report → Understanding → Classification → Impact Assessment → Prioritization → Routing.
- Use progress indicators, sequential cards or structured information reveals.
- Do not simulate hidden chain-of-thought or expose internal model reasoning.

Interaction Motion

- Cards: subtle elevation/translation/highlight.
- Forms: clear focus, validation and submission feedback.
- Status: visually confirm transitions and timeline updates.
- Maps: marker appearance, selection, focus and filter transitions.
- Charts: short informative entrance animations.

Timing Guidance

- Fast: 100–180 ms.
- Standard: 200–350 ms.
- Emphasis: 350–600 ms.
- Cinematic: 600–1200 ms and reserved for showcase moments.

- Use ease-out for entrances, ease-in for exits and smooth easing for large transitions.

Accessibility & Performance

- Respect reduced-motion preferences.
- On low-power devices simplify or disable heavy 3D.
- Use lazy loading, compressed assets, reduced particle counts and GPU-friendly transforms.
- 3D failure must fall back to simplified 3D, static visual or core UI.

WOW Moment

- Recommended story: From Citizen Signal to Community Intelligence.
- Example: citizen report → AI analysis → priority/department/safety relevance → admin assignment → status update → real issue appears in community intelligence.
- The moment must use real application data in the final demo.

Acceptance Criteria

- Hero experience defined; 3D purpose defined; AI analysis motion defined; issue/timeline/dashboard/map motion defined; reduced-motion behavior defined; mobile behavior defined; performance and fallback strategies defined; MVP motion priorities defined.

Decision Log

- D03-001 Motion communicates meaning, not decoration.
- D03-002 3D is optional for core functionality.
- D03-003 Core UX survives 3D failure.
- D03-004 AI animation never simulates hidden reasoning.
- D03-005 Reduced motion is required.
- D03-006 Advanced 3D cannot compromise MVP reliability.
- D03-007 Community intelligence visuals use real data.